

Arm's Length or Arm's Reach?

Transfer pricing and market foreclosure
in multi-club ownership networks

*A step-by-step guide to the thesis, with a detailed method
for building the player valuation (arm's-length price) model*

What this document contains

#	Section	What you will find
1	The thesis in one page	Research question, positioning against Quansah & Breuer (2025), hypotheses
2	Roadmap overview	The 8 phases, their outputs and the hours they require
3	Phases in detail	Goal, tasks, what you need, output and traps for every phase
4	Deep dive: the valuation model	Sample, target, features, leakage, models, validation, residuals, code
5	From residuals to answers	How to test H1-H4 and measure foreclosure
6	Timeline, risks, references	Month-by-month plan, risk register, reading list

1. The thesis in one page

Research question

Main question

Do multi-club ownership (MCO) groups trade players internally at prices that deviate from market value — and if so, when, why, and at whose expense?

The starting point: what is already known

Quansah & Breuer (2025, *European Sport Management Quarterly*) show that MCO groups use more loans internally and that intra-group transfer fees are higher, **on average**, than comparable transfers (propensity score matching on Transfermarkt data, 2017–2022). The fee result rests on only 44 intra-group paid transfers, the treatment is a simple yes/no dummy, and the authors call for future research using **clubs' financial data**, leaving the *why* open.

What this thesis adds

Dimension	Quansah & Breuer (2025)	This thesis
Question	Do intra-group fees differ on average?	When do they differ, why, and who bears the cost?
Price benchmark	Matching on Transfermarkt market value	Own machine-learning arm's-length price model, trained only on independent transfers, using pre-transfer information
Direction	Not distinguished (buyer and seller pooled)	Feeder → parent vs parent → feeder; round trips
Financial data	None (listed as future research)	Club-season financial pressure index
Consequences	Not studied	Market foreclosure; losses for sell-on holders, training clubs, rival clubs
Period	2017–2022	2014–2026 (post-BlueCo/Strasbourg, new UEFA rules)

Hypotheses (stated before looking at the results)

	Hypothesis	What the data should show if it is true
H1	Financial pressure (profit shifting)	Positive price gap when the seller is close to breaching financial rules; clustering before reporting deadlines (e.g. 30 June)
H2	Information advantage	Intra-group prices are closer to later realised value than prices between independent clubs
H3	Development strategy	Low prices for young players moving down the network, followed by cheap returns (round trips)
H4	Regulatory differences	Gaps linked to differences in national rules between the two clubs' countries, independent of pressure
H5	Market foreclosure	Breakout players at feeder clubs move to the parent more often, and at lower prices, than comparable players at independent clubs

Every outcome is a result: support for H1 is a regulatory finding; support for H2/H3 shows prices are legitimate; a null result under solid methodology is informative too.

2. Roadmap overview

The project runs in eight phases. Hours are the “medium” estimate (about 545 hours in total, roughly 7–9 months at 15–20 hours per week). Phases 2 and 3 are the heart of the original contribution; phase 3 is detailed in Section 4.

Phase	Goal	Main output	Hours
0. Set-up	Validate the topic, set up tools	Signed-off proposal, project repository	10
1. Literature review	Know what is known and where the gap is	Annotated bibliography, chapter 2 draft	60
2. Data building	Assemble the four datasets and merge them	One clean transfer-level table	155
3. Valuation model	Predict the arm’s-length fee of any transfer	Validated model + price gap for every transfer	50
4. Stage 1: measurement	Describe where and how prices deviate	Descriptive chapter, figures, matched comparisons	35
5. Stage 2: explanation	Test H1–H4	Regression tables, robustness checks	45
6. Stage 3: consequences	Measure foreclosure and third-party losses	Foreclosure estimates, loss estimates	45
7. Writing & defence	Turn results into a 40–50 page thesis	Final thesis, slides, (optional) paper version	155

Tools you will need throughout

- **Python** (pandas, numpy, scikit-learn, LightGBM or XGBoost, statsmodels, shap, optuna, matplotlib/seaborn, networkx). R is fine too if you prefer it for econometrics.
- **Environment:** VS Code or Jupyter, a Git repository (GitHub, private) so every step is reproducible, Google Colab if your laptop is slow.
- **Reference manager:** Zotero (free) with a Word or LaTeX plug-in. Start using it on day one.
- **A research log:** one document where you write down every decision (sample rules, thresholds, exclusions) with the date. Your methodology chapter will be written from it.

3. The phases in detail

Phase 0 — Set-up and validation (October)

Goal	Get the topic approved and avoid surprises later.
Tasks	Write a 2-page proposal (question, gap, data, method). Meet the supervisor. Ask EDHEC library about access to Orbis/Amadeus (company financials), Wyscout or Opta, and Taylor & Francis journals. Optionally e-mail Tommy Quansah (University of Lausanne) to present your extension.
You need	This document, the Quansah & Breuer paper, a 1-page summary of the planned data sources.
Output	Approved proposal; list of databases you have access to.

Phase 1 — Literature review (October–November)

Goal	Show precisely what is known and where your contribution sits.
Tasks	Read and summarise four bodies of literature: (1) multi-club ownership; (2) determinants of transfer fees and market values; (3) transfer pricing, related-party transactions and earnings management in football; (4) vertical integration and market foreclosure. Use the reference lists of Quansah & Breuer (2025) and of the 2026 “myth, expectation and evidence” paper to snowball.
You need	EDHEC library access, Google Scholar alerts on “multi-club ownership”, Zotero.
Output	Annotated bibliography (one paragraph per source: question, data, method, result, limit) and a draft of chapter 2 ending with your hypotheses.
Trap	Summarising papers one after the other. Organise the chapter by <i>ideas</i> (what is known about prices, about motives, about harm), not by author.

Phase 2 — Building the datasets (November–January)

This is the longest phase and your main original contribution. You build four datasets and merge them into one table where each row is a transfer.

2a. Ownership registry (built by hand, ~50 hours)

A table listing every MCO group, every club in it, the stake held and the dates. This does not exist publicly in usable form, so it is a contribution in itself.

Field	Example	Why it matters
group_name	BlueCo	Identifies the network
club_name / club_id	RC Strasbourg / Transfermarkt id	Join key with transfer data
stake_pct	99	Distinguishes control from minority investment
control_type	majority / significant influence / minority	Lets you test whether results depend on the strength of control
role	flagship / feeder / partner	Needed for direction (feeder → parent)
date_start, date_end	2023-06-xx, open	A transfer is “intra-group” only if both clubs were in the group on the transfer date
source	URL of press release / report	Traceability for the jury

Sources: UEFA European Club Finance and Investment Landscape reports, CIES club ownership report (2024), club and group websites, press archives, Wikipedia pages of each group as a *starting point only*.

Define a written rule for inclusion (for example: a stake of 30% or more, or board control, consistent with UEFA's notion of decisive influence).

2b. Transfers, players and market values (~30 hours)

- **transfermarkt-datasets** (GitHub project by dcaribou, also on Kaggle): clean tables of players, clubs, competitions, appearances, market valuations over time and transfers. Check the current schema and coverage first; complete missing transfer history by collecting Transfermarkt pages yourself if needed, respecting the site's terms.
- Keep for every transfer: date, player, selling club, buying club, fee (and whether it is disclosed), type (permanent, loan, loan with option or obligation, free), and the player's full valuation history.

2c. Performance data (~15 hours)

Minutes, appearances, goals and assists by season come from Transfermarkt appearances. For big-five leagues you can enrich with Understat (expected goals) or FBref. Performance must always be measured **before** the transfer (previous season, or the last 365 days).

2d. Financial pressure data (~60 hours)

Indicator	Source	Coverage
Pre-tax profit/loss, revenue, wages, equity	Companies House (UK, free, with API); Orbis via EDHEC for other countries; club annual reports	Very good in England, partial elsewhere
Distance to the loss limit (e.g. Premier League PSR, EFL rules)	Computed from accounts over the rolling assessment period	England
Sanctions and points deductions for financial breaches	League decisions, press	England, Italy, France
UEFA settlement agreements / financial sanctions	UEFA CFCB decisions (published on uefa.com)	Clubs in UEFA competitions
DNCG measures (salary caps, transfer restrictions)	DNCG decisions and annual report (LFP)	France

Combine these into a club-season **financial pressure index**: for example a dummy for “under sanction or formal supervision”, a continuous “distance to the loss limit” where available, and the club's loss as a share of revenue. Use England as the core sample for H1 if data elsewhere is too thin.

2e. Merging and cleaning (~25 hours)

- Match club names across sources with a mapping table (one Transfermarkt id per club). Never match on raw names.
- Convert all fees to euros at the transfer date and express them in constant money (see Section 4.3).
- Flag each transfer: **intra-group**, **inter-group** (both clubs in MCOs but different groups), **MCO-to-independent**, **independent**.
- Document every exclusion with counts (“12,431 transfers → 9,870 after removing undisclosed fees”). This becomes the sample-construction table in your methodology chapter.

4. Deep dive: building the valuation model

The idea in one sentence

The model learns **what independent clubs normally pay** for a player with given characteristics. Applied to an intra-group transfer, it gives the price two independent clubs would have agreed on — the **arm's-length price**. The gap between the actual fee and that prediction is your measure of mispricing.

4.1 The training sample: arm's-length transfers only

The model must never learn from the transactions you want to test, otherwise it would learn the very distortions you are trying to detect.

Keep in the training sample	Exclude from the training sample
Permanent transfers with a disclosed fee greater than zero	All intra-group and inter-group MCO transfers (they are scored later)
Both clubs independent of each other on the transfer date	Free transfers, swaps, undisclosed fees (no observable price)
Transfers from 2014 onwards (enough history for valuations)	Pure loans; treat loans with an obligation to buy separately (fee = loan fee + obligation)
	Obvious data errors (fee of €1, duplicates, fees 50× the market value without explanation — check by hand before deleting)

Expect several thousand to a few tens of thousands of training transfers, depending on coverage. That is plenty for gradient boosting.

4.2 The target variable: log of the fee

Predict **log(fee)** rather than the fee itself. Fees are extremely skewed (a few €100m deals and thousands of small ones); the log scale stops the model from being dominated by a handful of mega-transfers, and errors become **percentages**, which is what you want to compare: being €2m off on a €4m player matters far more than on a €100m player.

The residual $r = \log(\text{actual fee}) - \log(\text{predicted fee})$ reads directly as a relative gap: $r = 0.10$ means about 10% above the arm's-length price, $r = 0.69$ means about double, $r = -0.69$ means about half.

4.3 Features: what determines a transfer fee

Base the feature list on the transfer-fee literature (see the systematic review by Franceschi et al., 2023, in *Journal of Economic Surveys*). Group them as follows:

Group	Features	Comment
Player	Age, age squared, position, height, preferred foot, nationality region, international caps	Age is the strongest driver after value; the effect is non-linear
Contract	Months remaining on contract at the transfer date	Crucial: a player with 6 months left is cheap regardless of quality
Performance	Minutes share last season, appearances, goals and assists per 90, league level where minutes were played, trend versus previous season	Always measured before the transfer

Group	Features	Comment
Market signal	Last Transfermarkt valuation before the transfer, and its change over the previous 12 months	Powerful but crowdsourced; see leakage rules and the robustness version below
Selling club	League, league strength (UEFA country coefficient), club ranking, club revenue band	Big selling clubs negotiate higher prices
Buying club	League, league strength, revenue band, Premier League dummy	Premier League buyers pay a well-documented premium
Transaction	Season, summer or winter window, deadline day, same league or not, same country or not	Captures urgency and market conditions
Market level	Total spending in the buying league that window, or season fixed effects	Corrects for fee inflation over time

Leakage: the rule that makes or breaks the model

Every feature must be information that was available **before the transfer was agreed**. Transfermarkt often updates a player's value *after* a transfer, partly based on the fee. If you use that updated value, the model "sees" the answer and every residual shrinks towards zero.

Rule: use the most recent valuation dated **at least 30 days before** the transfer date; use performance from the previous season or the 365 days before; never use anything measured after the transfer.

Two versions of the model

- **Model A (main):** all features, including the pre-transfer Transfermarkt valuation. Most accurate.
- **Model B (robustness):** same features **without** any Transfermarkt valuation (fundamentals only). Less accurate but immune to the crowdsourcing bias that Quansah & Breuer acknowledge, which matters precisely for feeder clubs in less-watched leagues. If your conclusions hold with both models, they are much harder to attack.

Should buyer characteristics be included?

Yes, but understand what it means: the prediction becomes "what an *independent* buyer of this type (league, wealth) would pay". That is the right comparison for intra-group transfers, because a Premier League parent club would also pay a Premier League premium on the open market. Report one version without buyer features as a sensitivity check.

4.4 Which algorithms

Model	Role in the thesis	Notes
Naive benchmark: $\log(\text{fee}) = \log(\text{pre-transfer value})$	The bar to beat	If your model is not clearly better than this, it adds nothing
Linear hedonic regression (OLS)	Interpretable baseline linked to the economics literature	Coefficients can be compared with published studies
Gradient boosting (LightGBM or XGBoost)	Main model	Handles non-linearities and interactions (age $\times$ contract, position $\times$ league) automatically
CatBoost (optional)	Alternative	Handles categorical variables (league, nationality) natively

4.5 Validation: proving the model is reliable

- **Temporal split, not random.** Train on 2014–2022, tune on 2023, test on 2024–2026. This mimics real prediction and avoids the future leaking into the past. A random split overstates accuracy.
- **Group by player.** When you use cross-validation, keep all transfers of the same player in the same fold (GroupKFold), otherwise the model memorises players.
- **Metrics:** RMSE and MAE on $\log(\text{fee})$, median absolute percentage error on the fee, R^2 . Report them for the naive benchmark, OLS and boosting side by side.
- **Calibration plot:** predicted vs actual $\log$ fee on the test set. Points should follow the 45° line across the whole range, not only for cheap players.
- **Segment errors:** check accuracy by league, age group and fee band. If the model is much worse for small leagues, say so: that is exactly where feeder clubs sit.
- **Tuning:** optuna with time-ordered folds (learning rate, number of leaves, minimum samples per leaf, regularisation). Keep it simple; tuning gains are usually small.

4.6 Interpretability with SHAP

SHAP values show how much each feature pushed a given prediction up or down. Use them to (1) show the jury the model learned sensible economics (fees fall with age after 24–25, rise with contract length, Premier League premium, etc.) and compare with published findings; (2) explain individual cases in your results chapter, e.g. “the model priced this player at €12m mainly because of his age and 4 years of contract; the group paid €25m”.

4.7 The critical step: out-of-fold predictions for the control group

Why this matters

The model has **seen** the independent transfers during training, so its errors on them are artificially small. If you compare intra-group residuals (never seen) with in-sample independent residuals (seen), intra-group transfers will look abnormal even if they are not. Solution (cross-fitting): split the independent transfers into K folds; predict each fold with a model trained on the other $K-1$ folds. Every independent transfer then gets an **out-of-fold** prediction, exactly comparable to the predictions for intra-group transfers.

4.8 Uncertainty: when is a gap “abnormal”?

A single prediction is not enough: a 20% gap can be noise. Estimate a prediction interval for each transfer, either with quantile regression (LightGBM with a quantile objective, e.g. 5th and 95th percentiles) or with **conformal prediction**, which gives intervals with a guaranteed coverage rate. A transfer whose actual fee lies outside its 90% interval is flagged as an outlier. You can then compare the **share of outliers** among intra-group vs independent transfers: under no distortion, both should be close to 10%.

4.9 Pitfalls checklist

Pitfall	What to do
Undisclosed fees are not random (secret deals may be the strange ones)	Report how many intra-group transfers have undisclosed fees; test whether disclosure itself is less frequent within groups
Add-ons and bonuses (reported fee may include or exclude them)	Use the base fee consistently; note when sources report “up to” amounts
Loans with obligation to buy	Model separately or add loan fee + obligation as the effective price, with a dummy
Fee inflation over time	Season fixed effects or deflate fees by a transfer-market index
Currency conversion	Convert at the exchange rate of the transfer date
Small intra-group sample	Use all leagues and loans with fees; rely on intervals and non-parametric tests; be honest about power

4.10 Code skeleton

```
import numpy as np, pandas as pd, lightgbm as lgb, shap
from sklearn.model_selection import GroupKFold

df = pd.read_parquet("transfers_clean.parquet") # one row per transfer
FEATS = ["age", "age2", "position", "contract_months_left", "minutes_share_prev",
         "g_a_per90_prev", "log_mv_pre30d", "mv_change_12m", "sell_league_coef",
         "buy_league_coef", "buy_is_epl", "window_winter", "deadline_day", "same_league", "season"]
CAT = ["position"]

indep = df[df.flag == "independent"].copy() # arm's-length training set
mco = df[df.flag == "intra-group"].copy() # transactions to score
y = np.log(indep.fee_eur_real)

params = dict(objective="regression", learning_rate=0.03, num_leaves=31,
              min_data_in_leaf=40, feature_fraction=0.8, lambda_l2=1.0, verbose=-1)

# 1) Out-of-fold predictions for independent transfers (cross-fitting, grouped by player)
indep["pred_log"] = np.nan
for tr, te in GroupKFold(n_splits=5).split(indep, y, groups=indep.player_id):
    m = lgb.train(params, lgb.Dataset(indep.iloc[tr][FEATS], y.iloc[tr], categorical_feature=CAT),
                  num_boost_round=1500)
    indep.iloc[te, indep.columns.get_loc("pred_log")] = m.predict(indep.iloc[te][FEATS])

# 2) Full model to score intra-group transfers
full = lgb.train(params, lgb.Dataset(indep[FEATS], y, categorical_feature=CAT), num_boost_round=1500)
mco["pred_log"] = full.predict(mco[FEATS])

# 3) Price gaps (residuals): ~ % over/under the arm's-length price
for d in (indep, mco):
    d["gap"] = np.log(d.fee_eur_real) - d.pred_log

# 4) Explain the model
shap_values = shap.TreeExplainer(full).shap_values(indep[FEATS])
```

Note: the temporal train/test split of Section 4.5 is used to **evaluate** the model and choose its settings; the cross-fitting above is used to **produce the gaps** analysed in Stages 1-3. Both are needed.

5. From price gaps to answers

Phase 4 — Stage 1: measuring the discrepancies

- **Distribution comparison:** compare the gaps of intra-group transfers with those of independent transfers (means, medians, variance, share outside the 90% interval). Use non-parametric tests (Mann-Whitney, Kolmogorov-Smirnov) given the small intra-group sample.
- **Matched comparison:** propensity score matching (as Quansah & Breuer) but on your gaps, reporting covariate balance (standardised differences below 0.1) — a direct improvement on their Table 6.
- **Direction:** split by feeder → parent, parent → feeder, between sister clubs. Averaging directions can hide offsetting effects.
- **Patterns:** detect round trips (player leaves club A and returns to A via a sister club) and network flows (a graph of player movements with networkx).
- **Case studies:** 3–4 emblematic transfers explained with SHAP (e.g. Strasbourg → Chelsea moves).

Phase 5 — Stage 2: testing H1-H4

Core specification, estimated on intra-group transfers:

```
gap_i = a + b1*Pressure_seller + b2*Pressure_buyer + b3*NearDeadline_i
        + b4*Young_and_down_i + b5*RegDiff_i + controls + group FE + season FE + e_i
```

- **H1** predicts $b1 > 0$ (overpricing when the seller is under pressure) and $b3 > 0$ for sales just before the reporting deadline.
- **Placebo test:** run the same regression on *independent* transfers by clubs under pressure. Distressed clubs may sell cheaply (fire sales), so the pressure effect should be zero or negative there. A positive effect only within groups is strong evidence for H1.
- **H2:** compare how well intra-group prices predict the player's value two years later, versus independent prices.
- **H3:** interaction of age and direction (young players moving down the network) and the frequency of cheap returns.
- **H4:** dummies for pairs of countries with different financial rules.
- Use robust standard errors clustered by group; report results for Model A and Model B gaps.

Phase 6 — Stage 3: consequences

Market foreclosure (the Chelsea-Strasbourg question)

- **Define breakout players:** players at any club whose market value rises into the top decile of growth within two seasons (or a performance-based rule for Model B).
- **Compare** breakout players at feeder clubs with breakout players at comparable independent clubs (same league, age, value): probability of moving to the parent club vs to the open market (logit or survival model), and the price gap when they move.
- **Foreclosure is supported** if feeder-club talents move internally more often and at negative gaps, i.e. outside clubs never get to bid at market price.

Third-party losses

- **Solidarity payments:** for international transfers, FIFA redistributes up to 5% of the fee to the player's training clubs; estimate the shortfall as the solidarity share $\times$ (predicted – actual fee) when the gap is negative.
- **Sell-on clauses:** where a previous club's sell-on percentage is reported in the press, estimate its loss the same way (case-study level, since clauses are rarely public).

Phase 7 — Writing and defence

Chapter	Pages	Start writing
1. Introduction (context, question, contribution)	3–4	Last
2. Literature review and hypotheses	8–10	November
3. Institutional background (MCO, UEFA Article 5, financial rules)	4–5	December
4. Data and methodology (incl. the valuation model)	8–10	January–February
5. Results: measurement, explanation, consequences	10–12	March–April
6. Discussion and implications for regulators and clubs	4–5	April–May
7. Conclusion and limitations	2–3	May

6. Timeline, risks and reading list

Month-by-month plan (assuming submission around June 2027)

Month	Main work	Milestone
Oct 2026	Proposal, supervisor meeting, database access, start literature	Topic approved
Nov 2026	Literature review; ownership registry (top 15 groups); download transfer data	Annotated bibliography
Dec 2026	Finish ownership registry; performance data; start financial data (England)	Registry v1 complete
Jan 2027	Financial data; merging and cleaning; sample-construction table	Clean master table
Feb 2027	Valuation model: features, benchmarks, temporal validation, SHAP, cross-fitting	Validated model + gaps
Mar 2027	Stage 1 measurement; Stage 2 regressions and placebo tests	Results tables v1
Apr 2027	Stage 3 foreclosure and losses; robustness (Model B)	All results
May 2027	Writing chapters 5–7, full draft to supervisor	Full draft
Jun 2027	Revisions, formatting, defence slides; optional 20-page paper version	Submission

Risk register

Risk	Likelihood	Mitigation
Too few intra-group transfers with disclosed fees	High	All leagues, 2014–2026, loans with fees/obligations; descriptive and foreclosure parts carry more weight; honest power discussion
Financial data unavailable outside England	Medium	England as core H1 sample; sanctions/DNCG dummies elsewhere; ask for Orbis access
Model not much better than Transfermarkt value	Medium	Still useful: Model B and intervals add value; report it transparently
Leakage inflates accuracy	Medium	30-day rule on valuations, temporal split, cross-fitting
Null results	Medium	Pre-stated competing hypotheses make a null result informative
Ownership dates uncertain	Medium	Record sources; sensitivity analysis shifting dates by ± 3 months

Key reading list

- Quansah, T. K. & Breuer, C. (2025). Multi-club ownerships (MCOs): a critical analysis of transfer dynamics and sports integrity. *European Sport Management Quarterly*. doi:10.1080/16184742.2025.2474581
- Multi-club ownership and sporting success: myth, expectation, and evidence (2026). *European Sport Management Quarterly*. doi:10.1080/16184742.2026.2656332
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- OECD Transfer Pricing Guidelines for Multinational Enterprises and Tax Administrations (the arm's-length principle).
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- Lundberg, S. & Lee, S.-I. (2017). A unified approach to interpreting model predictions (SHAP). *NeurIPS*.